

Aboobaker Siddique

AI Application Developer | Python Backend Developer | Generative AI

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PROFESSIONAL SUMMARY

B.Tech Electronics & Communication Engineering graduate transitioning into software and AI engineering, with a strong Python backend foundation built through FastAPI and REST API development, SQLAlchemy/SQLite data modeling, and JWT/bcrypt-based authentication. Experienced in building Generative AI applications using the Gemini API, including prompt engineering, structured output generation, and LLM evaluation. Backed by hands-on project work rather than coursework alone, with an additional engineering strength in embedded systems and hardware prototyping. Currently focused on building production-ready AI applications and expanding into RAG systems and AI agents.

SKILLS

- Languages:** Python, SQL
- Backend:** FastAPI, REST APIs, SQLAlchemy, SQLite, Pydantic, CRUD, API Validation, Swagger/OpenAPI, Postman
- Authentication & Security:** JWT, bcrypt, Password Hashing, Protected Routes, Environment Variables
- Generative AI:** Gemini API, LLM Fundamentals, LLM Integration, Structured Outputs, Tokens & Context Windows, LLM Evaluation
- Prompt Engineering:** Zero-shot & Few-shot Prompting, Role Prompting, Reusable Prompt Templates, Output Constraints, Prompt Injection Awareness
- Tools:** Git, GitHub, VS Code, Postman, Swagger UI, Figma, Arduino IDE
- Embedded Systems:** ESP32, Arduino, Sensor Integration, Firmware Development, Circuit Design, Hardware Prototyping

PROJECTS

Siddique-AI | AI Application

- Personal AI application project; GitHub: AboobakerSiddique/Siddique-AI (see repository for full technical details).

AI Email Generator | Python, Google Gemini API, Prompt Engineering

- Built an LLM-powered email generation application using the Google Gemini API with reusable prompt templates incorporating role, context, and output-formatting constraints.
- Implemented structured output generation for email subject and body content, paired with input validation and API error/quota handling.
- Created test cases to evaluate output correctness, completeness, and tone, and tested prompt robustness against missing information, conflicting instructions, and basic prompt-injection attempts.

Secure Task Management API | Python, FastAPI, SQLAlchemy, SQLite, JWT, bcrypt, Pydantic

- Built a secure REST API with user registration, login, JWT-based authentication, and protected endpoints.
- Implemented bcrypt password hashing and CRUD operations for users and tasks, with input validation and unique email handling via Pydantic schemas.
- Documented and tested endpoints with Swagger/OpenAPI, and used environment variables to manage secrets and configuration.

dripVault – AI-Powered Wardrobe Management | Next.js, TypeScript, React, Tailwind CSS, Supabase, Gemini API

- Built a digital wardrobe management web app with AI-powered outfit generation using the Gemini API, user authentication, and Supabase-backed image storage, delivered through a responsive dashboard UI.

Embedded Systems Projects | Arduino, ESP32, Sensors, Firmware Development

- Hazard Guardian and RescueBot: Designed a multifunctional rescue robot integrating multiple sensors with a microcontroller to detect fire and gas leaks and trigger emergency alerts for disaster-management scenarios.
- Breathomicser: Developed a portable, ESP32-based breath-monitoring prototype using gas sensors and an OLED display for experimental health-screening applications.

EXPERIENCE

UI Designer, Lofilute

Jan 2024 – Nov 2024

- Designed responsive web and mobile interfaces, creating wireframes and high-fidelity prototypes in Figma for user-centered digital experiences.

Graphic Designer, Divegrid

Dec 2024 – Feb 2025

- Designed digital marketing and branding assets, collaborating with cross-functional teams to deliver visual design solutions.

Sales Staff, Planet Jewel

Jul 2024 – Sep 2024

- Assisted customers and handled sales operations in a customer-facing retail environment.

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering

University College of Engineering, Kariavattom | 2022 – 2026 | Languages: English, Malayalam